

Why Students Leave the LMS: Executive Function Demands and Self-Regulated Learning in Software Engineering Courses

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Abstract

Executive functions (EF)—including planning/organization, task initiation, working-memory management, cognitive flexibility, and help-seeking regulation—are foundational to engineering education and core software engineering practices. Yet little is known about how learning management systems (LMSs) support students who experience EF challenges in day-to-day coursework. We conducted an exploratory qualitative study using semi-structured interviews with seven engineering students both neurodivergent (ND) and neurotypical—who self-reported EF difficulties, and analyzed the data thematically through a self-regulated learning (SRL) and EF lens. Participants reported recurring EF-linked difficulties: (i) unclear or context-poor instructions prompting repeated re-reading and searches for external examples; (ii) a strong preference for brief visual/worked examples over dense text; (iii) peer-first help-seeking outside the LMS and limited use of discussion boards perceived as inconvenient; (iv) calendars valued and checklists helpful when used consistently (but usage was uneven); (v) email as the primary yet slow channel, with unease about formal messaging; (vi) growing reliance on AI for instant feedback and community sites (e.g., Stack Overflow) for coding questions. Overall, students regarded LMS tools as necessary for organization but insufficient for timely, low-effort clarification and support, leading to compensatory practices beyond the platform and increased cross-system switching. We synthesize an EF–SRL–LMS alignment that locates where EF load accumulates and derive a practice-oriented checklist. We argue for a more personalized, automated layer of in-platform support—adaptive, context-aware, and designed to reduce switching between multiple tools and platforms, to better sustain SRL across forethought, performance, and reflection while lowering EF load.

CCS Concepts

• **Applied computing** → **Learning management systems; Education**; *Computer-assisted instruction*.

Keywords

Executive function, Self-regulated learning, Learning management systems, Software engineering education, Neurodiversity

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1 Introduction

Executive functions (EF)—including planning/organization, task initiation, working-memory management, cognitive flexibility, and help-seeking regulation—are foundational for engineering study and core software engineering (SE) practices such as problem decomposition, debugging, coordination, and collaboration [16, 27, 31]. The role of EF in engineering education is significant, and deficits in EF are linked to lower academic performance across the general student population [5, 21, 23, 25]. The demands of post-secondary education in fields like engineering pose even greater challenges for individuals with poor EF compared to other academic disciplines [34].

Focused research on the day-to-day challenges faced by SE students and on how to support them effectively in virtual learning environments (VLEs) and learning management systems (LMSs) remains limited. The LMS is a course-administration platform (materials, announcements, assignments, grading), whereas a VLE denotes the broader set of online tools and spaces that support active participation and knowledge building; a VLE may subsume the LMS alongside interactive tools (e.g., webinar platforms, collaborative documents). In practice, the terms are sometimes used interchangeably [10]. While VLE/LMS platforms are positioned to promote self-regulated learning (SRL) by helping students plan, monitor, and reflect on their progress—capacities that are especially important in higher education [24]—we know far less about how well these tools support SRL under EF demands in software engineering (SE) coursework. In practice, EF domains such as planning/organization, task initiation, working memory management, cognitive flexibility, and help-seeking regulation can be strained by online study conditions like dense or ambiguous instructions, inconsistent use of calendars and announcements, fragmented help channels, and delayed responses. These conditions often require students (both neurodivergent and neurotypical students who struggle with EF) to switch between multiple tools and platforms to assemble context and obtain timely clarification, increasing EF load and disrupting SRL. Addressing this gap requires examining how such students utilize and perceive LMS tools and features for SRL, particularly in help-seeking contexts, to inform more inclusive instructional practices and platform use. We restrict the study to SE courses because that is where we teach and make platform decisions; this focus preserves ecological validity for recommendations about LMS/VLE use

(e.g., calendars, gradebooks, discussion boards, example provisioning). While SE is the focus of our inquiry, our findings may be more broadly applicable to similar programs such as other engineering disciplines.

EF drives routine SE work (planning, initiation, working memory, flexibility, help coordination) [18], yet most studies sort by diagnosis rather than shared EF challenges, obscuring how neurodivergent and EF-challenged neurotypical students experience LMS/VLEs for SRL and help-seeking. Using a transdiagnostic lens, we foreground EF to surface barriers common across contexts and to derive inclusive instructional and platform principles.

The goal is *to examine how engineering students who experience EF challenges—both neurodivergent and neurotypical—use and perceive LMS/VLE tools to self-regulate their learning in software engineering (SE) courses, with particular attention to help-seeking, and to identify barriers common across contexts that can inform inclusive instructional and platform practices.*

We conducted an exploratory qualitative interview study using semi-structured interviews with seven engineering students (neurodivergent and neurotypical) who self-reported EF challenges in SE coursework. Interviews were recorded, transcribed, anonymized, and analyzed using reflexive thematic analysis through a self-regulated learning SRL/EF lens.

The contributions of this paper are as follows:

- We report an interview-based characterization of LMS use under EF demands in SE courses. From semi-structured interviews with engineering students (neurodivergent and neurotypical with EF challenges), we identify recurring issues with clarity; a preference for brief visual/worked examples; limited forum use; reliance on announcements and calendars when kept current; slow email exchanges; and frequent recourse to peers and external tools (e.g., AI and community sites).
- We present a data-grounded EF–SRL–LMS alignment matrix that links EF domains to SRL phases and LMS touchpoints, and identifies where cognitive load accumulates and how cross-system switching affects help-seeking and self-regulation.
- We provide practice-oriented implications for inclusive LMS use in SE and principles for configuring instruction and LMS practices so that support is coherent, predictable, and largely within the LMS, thereby reducing cross-system switching and associated EF load.

2 Background

This section is organized into four parts. First, *Executive Functions (EF) and Relevance to Software Engineering* (2.1) defines EF and lists its core processes (e.g., planning/organization, working memory, cognitive flexibility, task initiation) and links them to key SE activities such as program comprehension, decomposition, tracing, and debugging. Second, *Self-Regulated Learning (SRL) and its relationship to EF* (2.2) outlines the SRL cycle (forethought, performance/monitoring, reflection) and explains how SRL phases overlap with EF processes. Third, *Students' experiences with LMS/VLEs* (2.3) synthesizes reported challenges in online course platforms, noting

gaps in the literature on day-to-day experiences of neurodivergent learners. Finally, *Responses to support and help with EF* (2.4) reviews supports discussed in the literature and highlights the limited evidence for EF-targeted, LMS specific supports.

2.1 Executive Functions and Relevance to Software Engineering

EF are control processes that allow learners to plan work, keep task information in mind, switch strategies, and initiate action [9]. In software engineering (SE) coursework, these processes are routinely required by core activities such as decomposing problems into units of work, tracing code paths, and debugging faults. In particular, *working memory*, *attention control*, and *cognitive flexibility* are essential to reading and modifying code, keeping multiple constraints in view, and shifting between abstraction levels during decomposition and debugging. Deficits in these areas can hinder efficient code tracing and error localization [8, 16, 27, 31]. Essential EF domains are related to SE activities and core programming skills:

- **Working memory and attention control.** Reading and modifying code requires maintaining variable states, call chains, and constraints while filtering irrelevant details. Tracing execution paths and aligning outputs with specifications require sustained attention and the capacity to keep intermediate representations active [8, 16, 27, 31].
- **Cognitive flexibility and inhibitory control.** Debugging and refactoring involve shifting between abstraction levels (e.g., algorithmic intent, data structures, control flow) and suppressing premature or irrelevant hypotheses. Flexibly switching strategies and perspectives supports efficient error localization and solution revision [8, 16, 27, 31].
- **Planning and organization.** Problem decomposition in assignments and projects depends on setting subgoals, ordering steps, and allocating effort across tasks (e.g., environment setup, test execution, incremental implementation). Clear step ordering and explicit criteria provide an external scaffold for planning and prioritization during multistep programming work [2, 11, 20].
- **Task initiation and sustained progress.** Moving from understanding a requirement to making the first code change reflects initiation. Short, explicit first actions (e.g., “run the failing test,” “replicate the error”) help translate intention into action and support continued focus on the task sequence [2, 11, 20].
- **Help-seeking regulation.** Determining when to consult documentation, search for examples, or ask questions depends on monitoring one's understanding and the perceived cost of interruption. Calibrated help-seeking supports timely clarification during programming and debugging without excessive disruption of task flow [17].

2.2 Self-Regulated Learning (SRL) and its relationship to EF

Self-Regulated Learning (SRL) is often described as a cyclical process with three phases: *forethought* (planning and goal setting), *performance* (strategy use and monitoring), and *self-reflection* (evaluation and adaptation) [36]. SRL provides a procedural complement to

EF. In the *forethought* phase, learners set goals, plan strategies, and allocate time activities that draw on EF for planning/organization and inhibitory control (e.g., committing to a first step rather than deferring). During *performance*, learners implement strategies, monitor progress, manage attention, and regulate help-seeking engaging EF processes of attention control and working memory while suppressing distractions. In *self-reflection*, learners evaluate outcomes, attribute causes, and adapt strategies, relying on cognitive flexibility to revise plans. Across phases, SRL encompasses goal setting, strategic planning, time management, self-monitoring, help-seeking regulation, and self-evaluation; extensive work links SRL to greater autonomy, persistence, and academic achievement [3, 4, 36].

2.3 Students' experiences with LMS/VLEs

Neurodivergent (ND) students (e.g., students with conditions such as ADHD, autism, and dyslexia) constitute a well-known group who often report EF challenges [4]. At the same time, they bring valued strengths for software engineering such as creativity, pattern recognition, and divergent thinking—the ability to explore many alternatives and perspectives when solving a problem [22, 32, 33, 35].

In software engineering courses that use LMS/VLEs, how students engage with course materials and workflows is crucial, yet the day-to-day LMS/VLE experiences of ND students are not well documented. The available evidence is sparse and fragmented; where studies are available, they frequently lack key context about the tools used (LMS or online learning tools), the learning modality (in-person, hybrid, online), and course characteristics. Students encounter practical and technical difficulties when interacting with LMS/VLEs such as confusing interfaces, problems uploading assignments, trouble entering provided passwords, missing spell-check or clickable links, and challenges filtering what is relevant [13, 29]. They also face overwhelming or poorly structured pages, particularly home pages that mix excessive or irrelevant content [13]. Locating and retrieving key materials can be time-consuming when information is buried in deep folder hierarchies or scattered across pages and tools [13, 29]. Guidance on how to navigate and use the system may be ineffective or poorly timed—for example, orientation that takes place long before students actually need the platform, or training that focuses on rarely used features [13]. In addition, some expected interactions are not well supported (e.g., phone numbers that are not clickable, tasks that require switching to other programs), which further complicates routine use [29].

2.4 Responses to support and help with EF

Evidence on EF-targeted supports within LMS/VLEs and for SE courses is limited; most reported adjustments are framed for ND learners in general rather than for specific EF difficulties. Commonly proposed measures include allowing learners to personalize what the system shows (e.g., views that surface only relevant items, use of color to signal state changes, selectable color schemes) and enabling smoother connections between tools so the experience feels seamless [13, 29]. Another approach emphasizes integrating complementary tools directly into the platform—such as artificial intelligence, learning analytics, text-to-speech readers, and spell-checkers to reduce friction in everyday tasks [4, 14, 29]. Organizing

course materials in a single, well-structured repository and separating social from academic information are also recommended to make essential resources easier to find and to avoid overload [13]. In line with Universal Design for Learning, multiple content formats (visual, audio, text) are encouraged [15]. Finally, involving ND learners in the design process (*co-design*) is reported to improve the fit of tools and programs, such as tailored strategies, peer mentoring, and study/life-skills modules [7]. Overall, these supports are described for ND students broadly rather than for discrete EF profiles; among the reviewed sources, only one explicitly foregrounds EF-related challenges and calls for co-design to develop assistive tools responsive to those needs [4]. Beyond platform level adjustments, there is also evidence for the efficacy of in-person EF focused interventions such as cognitive behavioral therapy (CBT), structured mentoring, and EF coaching. CBT typically combines weekly small-group sessions and individual mentoring to build time-management and organizational skills [1, 30]. However, it is unclear how these translate to a VLE.

3 Methodology

This section describes our approach to studying how engineering students who self-report EF difficulties engage with LMS tools in software engineering coursework.

3.1 Study Design and Context

We conducted an exploratory qualitative interview study to obtain detailed accounts of how engineering students who self-report EF difficulties use LMSs in software engineering (SE) coursework. Semi-structured interviews are well suited to revealing practices, pain points, and rationales that surveys may miss [28]. Planning and reporting followed established guidance for empirical SE studies and community standards [26].

Context. We focused on SE courses delivered in a *web-facilitated* mode on our institutional LMS, meaning primarily face-to-face courses that use the LMS for core logistics (e.g., materials, announcements, calendar, submissions, discussions) while instruction occurs in person [10, 19]. Grounding the study in this common LMS feature set gave participants a shared frame of reference and made workflows comparable across interviews, while still allowing differences in how individual instructors configured their courses.

3.2 Participants and Sampling

Eligibility criteria were (i) undergraduate or graduate engineering students who had completed or were enrolled in at least one SE course on our institutional LMS, and (ii) self-reported difficulty in one or more EF domains (e.g., planning/organization, task initiation, working memory, cognitive flexibility, help-seeking). We anonymized identities and refer to interviewees as I1–I7 (I = interviewee).

3.3 Recruitment and Setting

Participants were recruited via multiple channels: in-person outreach using a standard oral script; LMS course-homepage posts; posters shared through the Engineering Students' Society and Student Accessibility Services; and email invitations. Interested students reviewed study information and completed a short consent

Table 1: Interviewees and number of SE courses studied (self-reported)

Interviewee	# SE courses
I1	12
I2	10
I3	3
I4	4
I5	6
I6	2
I7	2

form. Recruitment materials described the study focus and were approved through the institutional ethics process. All interviews were conducted remotely by audio only. Interviews were semi-structured and focused on help-seeking behaviors and perceived usefulness of online tools and LMSes. The interviews lasted 35–55 minutes and were conducted by the first author. We recruited seven participants: four who self-identified as neurodivergent (three with ADHD and one autistic) and three who identified as neurotypical but reported difficulties in one or more EF domains (e.g., working memory, task initiation). Prior to the interview, participants were guided through the consent form, which included brief explanations of relevant EF domains, such as impulse control (e.g., resisting distractions and maintaining task priorities). The recruitment process was opportunistic (volunteer-based), not a polar design. Two additional students expressed interest and self-identified as neurodivergent but did not complete an interview; they are not included in the sample or analysis.

3.4 Data Collection and Analysis

Ethics and data handling. The study received institutional ethics approval. Participants gave informed consent and could pause or withdraw at any time. With permission, sessions were audio-recorded, transcribed, and de-identified at transcription. Audio and transcripts were stored on encrypted institutional drives with access restricted to the research team; each transcript and its audio share the same random study ID. The consent form allowed participants to optionally (i) indicate a neurodevelopmental condition from a list or select “prefer not to say,” and (ii) if they did not identify as neurodivergent, indicate any EF difficulties they experience. These attributes were used only in aggregate, stored separately from transcripts, and not linked to quoted IDs to reduce re-identification risk.

Data collection. Semi-structured interviews were conducted online. Each session lasted 25–35 minutes. The first author checked each transcript against the audio and wrote a short memo after the session.

Analysis. We used reflexive thematic analysis [6]. The first author read transcripts multiple times, noted initial ideas, and created codes. Some codes were informed by EF/SRL concepts; others were developed inductively from the data. Codes were compared across interviews and grouped into themes (T1–T5): T1: INSTRUCTION

CLARITY, T2: EXAMPLE-BASED LEARNING, T3: HELP CHANNELS, T4: TOOL CONSISTENCY, and T5: OFF-PLATFORM.

3.5 Data/Artifact Availability

Following recommendations on research data availability, we provide an anonymized snapshot of the analysis artifacts used in this study ¹.

4 Results

We analyzed seven semi-structured interviews with engineering students in SE coursework (neurodivergent and neurotypical) who self-reported EF difficulties. Participants described how they used LMS tools to self-regulate their learning and seek help. While LMS platforms provided essential organizational functions, students reported recurring challenges with *clarity*, *accessibility*, and *communication*, closely tied to EF demands in *planning/organization*, *task initiation*, *working memory*, *cognitive flexibility*, and *help-seeking regulation*. Below, we present the main Themes (T) with illustrative quotes and indicate (in parentheses) the EF domain(s) and SRL phase(s) primarily implicated.

4.1 How engineering students use and perceive LMS tools for SRL in SE course

- (1) T1: INSTRUCTION CLARITY (EF: working memory, task initiation; SRL: Plan/Do). Students described re-reading dense briefs and leaving the LMS to find clearer examples when task information lacked structure or context. “*The nature of slides ... sometimes not being annotated and notes sometimes not being thorough enough.*” (I2)
- (2) T2: EXAMPLE-BASED LEARNING (EF: working memory; SRL: Plan/Do). Participants consistently preferred brief, step-wise demonstrations over text-heavy pages. “*...for example, for a question. It provides the solution in steps and shows you how to do it properly.*” (I6)

4.2 Help-seeking choices shaped by effort and timeliness

- (1) T3: HELP CHANNELS (EF: help-seeking regulation; SRL: Do). Students used multiple channels to seek help and weighed effort against expected response time. Many consulted peers via short messages or voice notes rather than posting in LMS forums. Discussion boards were viewed as occasionally useful but more convenient if anonymous posting were available; email to instructors was common yet often slow and sometimes felt inappropriate for quick questions. Some students avoided formal or direct messaging altogether. “*...The best way to reach out (to instructors and teacher assistants (TAs) is through email. It’s a little too formal, I believe, for me to, like, send them a team’s message. And I often find email can take a while for them to respond. Or sometimes they don’t respond at all.*” (I1)

¹https://github.com/Shnaikat/Supplemental_Materials_For_ICSE_SEET_2026

4.3 Consistency and cross-platform switching under EF load

- (1) T4: TOOL CONSISTENCY (EF: planning/organization; SRL: Plan/Do). Students valued LMS calendar supported planning and progress tracking when instructors kept it current; inconsistent use made it difficult to monitor deadlines and upcoming tasks. “... One of my profs did use the calendar and he said like I use the calendar religiously. Any due date will be found in the calendar. I found that incredibly helpful.” (I1)
- (2) T5: OFF-PLATFORM (EF: cognitive flexibility, working memory; SRL: Plan/Do/Reflect). Students used external tools or resources such as AI tools (e.g., ChatGPT) to obtain feedback on their work and relied on community forums (e.g., Stack Overflow) for programming-specific questions. “I know I’m doing something wrong, but I can’t find where I’m doing it So I just take pictures of my work and then it tells me what step I messed up...” (I5)

4.4 Summary of patterns

Across themes, students regarded LMS tools as necessary for organization but insufficient for timely, low-effort clarification and support. As a result, they engaged in compensatory practices beyond the platform (peers, AI, community sites), increasing cross-system switching and the associated EF demands (working memory, cognitive flexibility, and help-seeking regulation).

5 Discussion

Our findings show that students who report EF challenges, including neurodivergent students and neurotypical students with EF difficulties, use LMS tools in ways constrained by *clarity*, *consistency*, *timeliness*, and *effort*. While neurodivergent students sometimes described these demands as more persistent or intense, the types of challenges were similar across groups, suggesting that the support strategies we propose address shared needs rather than group-specific differences. Below, we interpret the themes from Section 4, consolidate mechanisms via an EF–SRL–LMS alignment, and translate them into course/LMS practices.

5.1 Interpreting the themes: where EF load accumulates

Taken together, the themes indicate that EF costs concentrate at specific points in LMS-mediated coursework. Below we connect each theme, to the EF domains and SRL phases most affected, and we indicate the likely consequence for students’ self-regulation.

- (1) T1: INSTRUCTION CLARITY. When task instructions are unclear or lack context, *working memory* load increases and *task initiation* is delayed (SRL: Plan/Do). Students re-read, hold partial details in mind, and search for missing explanation and guidance, which elevates cognitive effort before any productive work begins.
- (2) T2: EXAMPLE-BASED LEARNING. Brief, stepwise, visual or worked examples reduce *working memory* demands by externalizing intermediate steps and expected outputs (SRL: Plan/Do). This lowers the cost of getting started and helps

students sustain progress without leaving the LMS to look for clarifying materials.

- (3) T3: HELP CHANNELS. Uncertain or slow help pathways increase the cost of *help-seeking regulation* (SRL: Do). Students weigh effort against expected response time; high-effort forums and delayed email replies push them toward peers or informal channels, which can fragment the workflow and reduce predictability.
- (4) T4: TOOL CONSISTENCY. Calendars and checklists support *planning/organization* and monitoring (SRL: Plan/Do/Reflect) when they are used reliably. EF load accumulates when updates are inconsistent: students must reconcile multiple sources, increasing tracking effort and the likelihood of missed or late actions. Consistent, in-LMS feedback and results (e.g., use of the grade book with clear next steps) reduce *working-memory* and *monitoring* demands during reflection by keeping outcomes and follow-up actions in one place.
- (5) T5: OFF-PLATFORM. Relying on external tools for explanations or coding support increases *cognitive flexibility* and *working memory* demands (SRL: Plan/Do). Switching between systems (e.g., LMS, AI, community sites) requires context reconstruction and item juggling, which raises the cost of maintaining task state and integrating feedback.

To summarize, EF load accumulates where instructions lack structure, help pathways are effortful or slow, course tools are inconsistently maintained, and students must switch across platforms to obtain examples or feedback. Aligning LMS practices with these pressure points (clearer instructions with an explicit first step, concise worked examples in-platform, a predictable help pathway, and consistent calendars/announcements) is likely to reduce initiation costs, lower working-memory burden, and support more stable SRL across Plan, Do, and Reflect.

5.2 Synthesize the Themes with the Literature Findings

In the literature findings, ND students reported practical and technical issues when interacting with the LMS/VLE, including confusing interfaces, assignment-upload problems, difficulty entering passwords, lack of spell-check or clickable links, and difficulty identifying relevant information. [13, 29]. In our data, this aligns with reports of inconsistent use of LMS tools, which raise EF demands on attention and working memory.

Furthermore, the students reported overwhelming or poorly structured pages that mixed excessive or irrelevant content, especially on course home pages. [13]. Our participants’ preference for brief, worked/visual examples and difficulty with dense, text-only materials, fits this theme by showing how overload amplifies EF load during day-to-day SE coursework.

Students struggled to locate or retrieve key items because information was buried in deep folder structures or scattered across pages and tools. [13, 29]. Our results echo this for inconsistent calendars/checklists and lacking feedback, which add planning and working-memory costs and slow reflection/next-step decisions.

Students encountered ineffective or poorly timed guidance on how to navigate and use the system (e.g., introductory VLE sessions

scheduled long before use; coverage of rarely used features) [13]. Our interviews extend this by showing that unclear or context-poor assignment instructions, and the absence of an explicit “first step” increase initiation costs and repeated re-reading and looking for other resources.

The literature also notes issues when ND students interact with VLEs. When the system expects user action, tasks can be hard to complete because supportive information is missing (e.g., non-clickable phone numbers; tasks that require switching to other software) [29]. We observe a related pattern—effortful or slow help pathways that push students to leave the LMS (email delays, low forum uptake), increasing cross-system switching and EF load (attention shifting, working-memory refresh).

5.3 Extensions beyond literature findings

Two recurring patterns in our interviews are only weakly identified in prior categorizations were: (i) *peer-first help* outside the LMS (personally or through short messages) as a lower-effort alternative when in-LMS help feels slow or high-friction, and (ii) *increasing reliance on AI tools* for instant feedback and quick exemplars. Both reduce immediate interpretation effort but increase cross-system switching, with implications for EF demands.

5.4 EF–SRL–LMS alignment matrix (EF domains, SRL phases and LMS touchpoints)

The alignment matrix (Table 2) summarizes where EF load accumulates across SRL phases and identifying the implicated LMS touchpoints. Additionally, it presents different supports to help the students to self-regulate their learning and reduce the potential EF load. Below we illustrate these supports: (i) Clear and coherent assignment prompts (Task brief or description) reduce working-memory demands and help students initiate tasks more quickly (Plan/Do); (ii) consistent calendars/announcements support planning/monitoring (Plan/Do); (iii) one explicit help pathway with stated reply windows reduces the regulatory cost of help-seeking (Do); (iv) minimizing cross-platform switching curbs cognitive-flexibility and working-memory load (Plan/Do/Reflect); and (v) centralized feedback enables efficient reflection and next-step decisions (Reflect).

5.5 Practice-oriented implications for inclusive LMS use in SE

To operationalize these supports, Table 3 provides a concise review checklist for configuring instruction and LMS practices for more inclusive learning environments.

Mapping the themes to the checklist calls for:

- **Structured assignment and course material descriptions for initiation:** Make task goals and first steps explicit; embed a short worked/visual example in-platform (T1–T2).
- **Standardize organizational conventions:** Use stable calendar/announcement conventions across offerings to reduce planning overhead (T4).
- **Unify help pathways:** Provide one visible in-LMS channel with typical response windows; avoid parallel, competing channels (T3–T4).

- **Minimize cross-system switching:** Keep essential explanations/examples inside the LMS; avoid required hops during Plan–Do–Reflect (T1–T5).
- **Centralize feedback and next steps:** Ensure outcomes and forward actions live in one in-LMS location to support reflection (T4).

5.6 Insights From an Introductory Python Course: EF–SRL–LMS Alignment and Implications for SE Teaching

Because all participants completed an introductory Python course, their experiences illuminate how foundational software engineering activities appear in early coursework. Even introductory programming requires students to interpret requirements, decompose problems, iteratively develop and debug code, document decisions, review examples or peers’ work, and manage multiple deadlines. These activities place sustained demands on planning, working memory, cognitive flexibility, and help-seeking across the SRL phases. Unlike many non-SE courses, students must coordinate multiple artifacts (e.g., code, tests, instructions, outputs) and reason across representations, making LMS structure and instructional design particularly consequential. The EF–SRL–LMS alignment matrix helps explain why SE courses require more intentional pedagogical and structural support, even at the introductory level.

In the **Plan** phase, students translate task descriptions into actionable steps and identify dependencies. Unclear goals, scattered updates, or missing first steps increase planning overhead and hinder task initiation. This effect is amplified in SE courses, where assignments often span multiple files or stages and early decisions shape later work. In the **Do** phase, students move between the LMS, code editors, test outputs, documentation, and external resources, increasing working-memory and cognitive-flexibility demands. Reliance on StackOverflow, commented examples, and code-explanation tools reflects the need for scaffolding when in-LMS resources lack clarity or immediacy. This pattern reflects a defining feature of SE learning, where understanding emerges through iterative testing, error interpretation, and comparison with exemplars rather than linear content consumption. In the **Reflect** phase, fragmented feedback across systems makes it difficult for students to evaluate progress and refine debugging and problem-solving approaches.

These SE-specific demands directly motivate the checklist recommendations. Clear task descriptions support initiation; standardized organization reduces planning load; unified help pathways clarify where to seek assistance; minimizing cross-system switching supports focus across the Plan–Do–Reflect cycle; and centralized feedback supports reflection. In SE courses, where learning depends on sustained engagement with complex, evolving artifacts, these choices enable core practices such as abstraction, systematic debugging, and iterative refinement. Together, they allow students to focus cognitive resources on developing SE competencies rather than managing logistical complexity.

As instructors use the LMS as the starting point for course tasks, students typically access assignments via the LMS/VLE. Instructors should place worked examples and key resources directly on the

Table 2: EF–SRL–LMS Alignment Matrix for SE Courses

EF domain	Plan (Forethought)	Do (Performance/Monitoring)	Reflect
Planning / organization	LMS touchpoints: Calendar, Announcements, Task brief. Supports: Stable due-date conventions; single location for changes. Risk: Inconsistent calendars; scattered updates. SE tasks often span multiple files or milestones, amplifying planning costs when schedules shift.	Supports: Visible “what’s next” outline within the course shell. Risk: Hidden dependencies; shifting requirements. Dependencies are common in programming assignments (e.g., function order, test readiness).	Supports: Clear location for grades/criteria and next actions. Risk: Feedback spread across tools. SE learning depends on integrating feedback into subsequent iterations.
Task initiation	LMS touchpoints: Task brief, Resources. Supports: Concise goal statement; explicit first step; brief worked/visual example. Risk: Unclear or context-poor instructions → repeated re-reading. SE assignments often lack a single obvious starting point.	Supports: Direct, <i>in-LMS</i> access to examples and resources. Risk: Leaving LMS to find examples and resources (peers/AI/Stack Overflow). Code comprehension commonly relies on examples and comparisons.	Supports: Short <i>in-LMS</i> prompt for “what I did / what to improve”. Risk: Reflection pushed to external tools.
Working memory	LMS touchpoints: Brief, Rubric. Supports: Chunked sections, headings, criteria summary. Risk: Long, dense pages without structure. Programming tasks already require tracking multiple states and variables.	Supports: Same-page access to instructions and the submission area. Risk: Switching tabs to recall details. Context switching disrupts code tracing and debugging.	Supports: Compact summary of feedback and criteria in one place. Risk: Feedback scattered across emails and forums.
Cognitive flexibility	LMS touchpoints: Navigation/layout. Supports: Consistent navigation and page structure across courses/terms (same menu labels and locations). Risk: Instructor-specific layouts and naming that diverge from the standard.	Supports: Minimize switching between multiple tools and platforms during tasks. Risk: LMS → email → forum → AI → back (context loss), and the need to verify AI-generated explanations (e.g., from ChatGPT) adds additional effort and disrupts task focus. SE work already requires frequent shifts between representations (code, output, documentation).	Supports: Unified view of outcomes and next steps within the LMS. Risk: Outcomes fragmented across systems.
Help-seeking regulation	LMS touchpoints: Help block/FAQ, Discussion board policy. Supports: One explicit pathway; stated response windows. Risk: Parallel/competing channels. SE learners often seek rapid clarification to unblock progress.	Supports: Timely replies; searchable prior answers; clear tone expectations. Risk: Forums perceived as inconvenient; email delays; discomfort with formal messaging → shift to peers/AI. External tools are frequently used to interpret errors or code behavior.	Supports: Allow anonymous posts in discussion forums, but include safeguards (e.g., review or reporting) to prevent inappropriate content. Closure indicators (“Answered/Resolved”) and a defined escalation pathway for unanswered posts. Risk: Unresolved threads with no closure indicator or escalation route.

Table 3: Course/LMS Review Checklist (yes/no)

Question	Evidence to check
Are all dates/changes presented in a single, consistent place?	Calendar consistency; announcement location and naming conventions.
Is the first step of each task explicit in the description?	One-sentence goal plus first action included with the task.
Can students access a short worked/visual example and related resources inside the LMS?	Example link placed near the brief/resources (no external hop required).
Is there one help channel with stated response windows?	“Where to ask” block visible; typical reply times; escalation path.
Can students complete Plan–Do–Reflect without leaving the LMS?	Count of external hops per task; centralized feedback and next steps.

assignment page (or in a single “Course Resources” hub) and reference them in each task. Keeping help where students start reduces off-platform searching and rebuilding the context.

Why: Concentrating resources on the assignment page lowers task-initiation effort and working-memory load during performance.

Risks addressed:

- **Task initiation (Plan/Do):** Unclear or context-poor instructions → repeated re-reading; leaving the LMS to find examples/resources.
- **Working memory (Plan/Do):** Long, dense pages without structure; switching tabs to recall details.

Recommendation 1

Put the worked example and key links on the assignment page (or a “Course Resources” hub) and point to them in each related task.

EF demands spike at predictable points: unclear or unstructured task briefs (Plan/Do) and inconsistent organization of dates/updates (Plan/Do). Address these issues at their source by making the task brief easy to scan and using the calendar as the single, reliable place for all deadlines and changes.

Why: Clear, chunked briefs and a single source of truth for dates reduce repeated re-reading, guesswork about dependencies, and off-platform searching.

Risks addressed:

- **Task initiation (Plan/Do):** Unclear or context-poor instructions → repeated re-reading; leaving the LMS to find examples/resources.
- **Planning/organization (Plan/Do):** Hidden dependencies; shifting requirements; inconsistent calendars; scattered updates.
- **Working memory (Plan/Do):** Long, dense pages without structure.

Recommendation 2

Begin each brief with a one-sentence goal, an explicit first step, and a small worked/visual example; use headings/bullets and list dependencies. Put all dates and changes in the LMS calendar and link back to the assignment page.

Provide one in-LMS help pathway with a posted reply window (e.g., 24–48h) and a clear escalation route; allow anonymous questions **with moderation**: posts are reviewed by a TA or instructor before going live, or there is a reporting/flagging system so inappropriate or off-topic content can be quickly removed. Keep prior answers searchable. Keep feedback, rubric/criteria, and “what to do next” in a single LMS location, and mark discussion threads “Answered/Resolved.”

Why: A single, predictable pathway lowers the effort of help-seeking and keeps reflection in one place; closure markers reduce uncertainty and unnecessary back-and-forth. Participants also discussed turning to AI tools such as ChatGPT when LMS-based help felt unclear or slow, using them to obtain alternative explanations or step-by-step reasoning. While this can lower working-memory demands and support task monitoring, students expressed concerns about accuracy and noted the added effort of verifying AI-generated responses. A unified and easily accessible in-LMS help pathway therefore reduces unnecessary shifts to external tools and supports more reliable, trustworthy help-seeking within the course environment.

Risks addressed:

- **Help-seeking regulation (Plan):** Parallel/competing channels.
- **Help-seeking regulation (Do):** Forums perceived as inconvenient; email delays; discomfort with formal messaging → shift to peers/external tools including ChatGPT, which students viewed as helpful for alternative explanations but risky to trust without additional verification).
- **Help-seeking regulation (Reflect):** Unresolved threads with no closure indicator or escalation route.
- **Working memory / Cognitive flexibility:** Outcomes fragmented across systems.
- **Anonymous option (Potential risk for the support):** Inappropriate content if not moderated.

Recommendation 3

Keep help and feedback in one predictable place in the LMS; avoid channel and tool switching during a task.

Organize the course so students can Plan, Do, and Reflect with minimal extra effort. In practice, that means showing the goal and first step up front (Plan), keeping instructions/resources and submission together (Do), and putting feedback, criteria, and “what to do next” in one place (Reflect).

Why: Small, consistent prompts at each phase lower task-initiation effort, reduce page-hopping, and make next steps obvious.

Risks addressed:

- **Planning/organization (Plan/Do):** Inconsistent calendars; scattered updates; hidden dependencies; shifting requirements.

- **Task initiation (Plan/Do):** Unclear or context-poor instructions → repeated re-reading; leaving the LMS to find examples/resources.
- **Working memory (Plan/Do):** Long, dense pages without structure; switching tabs to recall details.
- **Help-seeking regulation (Do/Reflect):** Parallel/competing channels; forum inconvenience and email delays → shift to peers/external tools; unresolved threads without closure/escalation.
- **Cognitive flexibility (Reflect):** Outcomes and feedback fragmented across systems.

Recommendation 4

Use a light-touch, EF-aware setup: *Plan* (goal, first step, dates/dependencies), *Do* (instructions + example + submission together), *Reflect* (feedback, criteria, and next actions in one place).

6 Conclusion

In this section we provide suggestions for future work, discuss the study's limitations, and summarize the paper.

6.1 Future Work

Our study provides an initial interview-based account of how EF demands shape LMS use in SE courses and introduces two synthesis artifacts: the EF–SRL–LMS alignment and a course/LMS checklist. While the findings are not intended to be broadly generalizable, they offer actionable guidance for supporting self-regulated learning among students who experience EF challenges, including neurodivergent learners, particularly in environments where LMSs and VLEs are increasingly integrated into everyday study practices. Given the small, institution-specific sample and the use of a single LMS configuration, the findings should be interpreted as illustrative rather than representative of all SE contexts.

Future work should replicate the study with larger and more diverse cohorts across institutions, course levels, and delivery modes, and should compare different LMS platforms and local course conventions to test how robust the themes are. Such replications would help establish the extent to which the observed EF–LMS interactions generalize beyond the present context and identify which supports are broadly effective versus context-dependent. In particular, cross-institutional studies using different LMSs and instructional norms would clarify which challenges stem from SE pedagogy itself versus local platform or course design choices.

Methods could be broadened through triangulation by combining interviews with LMS log data, timestamped communication records, and artifact analysis so that qualitative accounts and quantitative indicators can inform and check one another. The alignment and checklist should be turned into specific course changes and then evaluated in practice. Examples include clarifying briefs (goal + first step + a short worked example embedded in the LMS), providing a single in-LMS help pathway with stated reply windows, and centralizing feedback; these changes can be evaluated for effects on time to clarity, help-seeking, cross-system switching, and performance or engagement. Evaluations should also examine whether

these supports differentially benefit students with EF challenges compared to the broader student population. A light-touch, context-aware in-LMS assistant that offers just-in-time micro-supports and routes help within the LMS could be prototyped and compared with current practice on EF-related outcomes and help-seeking behavior. The supports in the matrix should also be operationalized in a reusable LMS course template that standardizes navigation and labels, pre-populates assignment pages (goal, inputs, first step, worked-example slot, single help block with reply windows, centralized feedback), and uses lightweight checks for required elements and consistent naming and due dates; the effects of this template should be evaluated. Testing such templates across multiple institutions and LMS platforms would further strengthen claims about transferability and generalizability. Longitudinal studies should follow cohorts across terms to test durability, including impacts on EF load, task initiation, course completion, and retention. Finally, precise reporting of context (course size, delivery mode, platform configuration, assignment types, response-time norms) and the development of reusable pattern catalogs or standards handbooks for EF-aware LMS practices would support synthesis across studies and speed adoption.

6.2 Limitations

We consider trustworthiness using Guba's (1981) dimensions of *credibility*, *transferability*, *dependability*, and *confirmability* [12].

6.2.1 Credibility. Credibility concerns the truth value of the findings. Our analysis draws on direct participant quotations and an interview guide explicitly aligned with EF/SRL domains (planning/organization, task initiation, working memory, cognitive flexibility, help-seeking), supporting a faithful reading of participants' accounts. Because the data are self-reported, they may include *recall errors* (e.g., generalization) and *social-desirability effects* (e.g., overstating desirable practices, underreporting reliance on peers/AI). We did not triangulate with trace data (e.g., LMS logs). Some interpretations may therefore reflect participants' perceptions rather than observed behavior under real task constraints. To aid interpretation, we summarize prior literature on LMS use by students experiencing EF challenges in the Background section (2) section, synthesize the findings of the literature with our results in the Discussion section (5) and draw on the first author's experience with student LMS practices to clarify terminology and workflows. These steps provide context for readers' judgment, but they do not replace triangulation with trace data (e.g., LMS logs). Additionally, our small sample size may affect the credibility of the findings. Despite a nine-month recruitment effort, participation was likely limited by social stigma among ND students and the heavy time demands on engineering students. Additionally, our small sample size may affect the credibility of the findings. Despite a nine-month recruitment effort, participation was likely limited by social stigma among ND students and the heavy time demands on engineering students. To mitigate low response rates, we employed multiple recruitment strategies over the full nine-month period, including repeated in-person outreach using a standard script, LMS course-homepage posts, and email invitations. When initial responses remained limited, we extended recruitment to additional channels, including the Engineering Students' Society and Student Accessibility Services,

to reach students who might otherwise be underrepresented. Although two additional students completed the consent form and expressed interest, they did not proceed to the interview stage; no reasons were provided, a pattern consistent with barriers reported in prior work, such as time constraints and social stigma associated with neurodiversity.

6.2.2 Transferability. Transferability addresses the applicability of findings to other contexts. The sample ($n=7$) comprises engineering students (neurodivergent and neurotypical) who self-reported EF difficulties and were studying SE courses delivered in a web-facilitated LMS modality; this purposive sampling fits the study purpose but is not representative of all SE students or institutions. LMS configurations and course conventions vary across programs and instructors; therefore, some patterns (e.g., calendar/announcement practices, help channels) may differ elsewhere. Because our teaching and platform decisions occur within SE courses, the study emphasizes SE settings; however, several mechanisms we describe (e.g., EF load associated with instructional clarity, calendaring/announcement practices, and help-seeking workflows) likely extend to other engineering domains that use similar LMS/VLE configurations. The transdiagnostic EF lens makes the findings relevant across diagnostic labels (e.g., ADHD, autism, dyslexia), but it may not capture nuances unique to particular conditions or to specific course formats (e.g., studio/lab, project-based, large lecture, fully online). LMS configurations and local teaching conventions also vary across programs. We therefore describe the course teaching modality and LMS context in the Methodology section (3), so the study should be treated as adaptable, not universal. Additionally, the EF–SRL–LMS alignment and checklist synthesize interview findings from our setting. They should be read as data-informed guidance, not as tested interventions. Demonstrating effectiveness will require follow-up evaluation in varied contexts and attention to cognitive equity among student groups.

6.2.3 Dependability. Dependability asks whether another researcher, using the same materials and steps, would reach broadly similar themes. We aimed for consistency by (i) using the same semi-structured interview guide for all participants, (ii) keeping a single codebook with brief definitions and example quotes, and (iii) applying the same routine to each transcript (first-pass coding, grouping into themes, and re-checking earlier transcripts after any revisions). We also noted when the guide or codebook changed and why, so we can ensure the reliability of the results.

6.2.4 Confirmability. Confirmability concerns whether the findings reflect the data rather than the researcher. We sought to limit researcher influence by: (i) maintaining an audit trail (dated interview-guide revisions, a versioned codebook with brief notes on edits, and a short decision log linking coding choices to emerging themes); (ii) keeping reflexive memos that recorded prior assumptions and how they were addressed; (iii) actively seeking and reporting disconfirming cases when refining themes; and (iv) anchoring claims in direct participant quotations. Together, these steps provide a traceable chain of evidence from interview excerpts to codes to themes. Even so, coding by a single analyst and the use of an a priori EF/SRL lens are constraints that may have foregrounded certain mechanisms (e.g., working-memory load, initiation costs).

6.3 Conclusion

This study examined how engineering students who report EF challenges—both neurodivergent and neurotypical—use learning management systems (LMSs/VLEs) to regulate their learning and seek help in software engineering (SE) courses. Using semi-structured interviews ($n=7$) and thematic analysis through an EF/SRL lens, we identified recurring barriers tied to EF demand: unclear or context-poor instructions that prompt re-reading and external searching; a strong preference for brief visual/worked examples; help-seeking shaped by effort and timeliness (forums seen as inconvenient; email primary but delayed; discomfort with formal messaging); organizational tools that help only when used consistently (gradebook, calendars); increasing reliance on AI and community sites for explanations and coding support; and the lack of feedback that complicates reflection and next-step decisions. Overall, students regarded LMS tools as necessary for organization but insufficient for timely, low-effort clarification and support, leading to compensatory practices beyond the platform and added EF load.

We contribute two synthesis artifacts derived from the interviews: (i) an **EF–SRL–LMS alignment matrix** that locates where EF demand accumulates across Plan/Do/Reflect phases and identifies implicated LMS touchpoints and present the needed supports, and (ii) a **course/LMS review checklist** that translates these supports into practical, in-platform actions to be followed when designing SE courses. Together, they motivate a light-touch, EF-aware configuration for SE courses: clear assignment briefs (goal + first step), consistent calendars/announcements, embedded worked-example files and resources inside the LMS, a single in-LMS help pathway with stated reply windows, and centralized feedback with next steps. These principles aim to reduce switching between multiple tools and platforms, thereby lowering working-memory and cognitive-flexibility demands while supporting self-regulated learning.

Our analysis is qualitative and context-specific; findings should be read as data-informed guidance rather than validated interventions. We encourage instructors and course teams to apply the checklist to their local LMS configurations and to report outcomes, and researchers to evaluate the alignment and checklist in varied SE contexts (e.g., different course sizes/modalities and LMS setups). Clear reporting of course context and terminology will also aid comparison across studies and help build a cumulative, evidence-based approach to accessible and inclusive software engineering education.

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